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MIL-P-85833(AS)
6 November 1989

MILITARY SPECIFICATION

PROTECTIVE ASSEMBLY, HELICOPTER AIRCREWMAN, CHEMICAL, BIOLOGICAL, RADIOLOGICAL

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of a protective assembly (PASS) to provide head, eye and respiratory protection for helicopter aircrewmen in a chemical, biological and radiological (CBR) environment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-6051	-	Electromagnetic Compatibility Requirements, Systems.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 53), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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SPECIFICATIONS

MILITARY (cont'd)

MIL-N-18307	-	Nomenclature & Identification for Aeronautical Systems Including Joint Electronic Type Designated Systems & Associated Support Systems.
MIL-H-46855	-	Human Engineering Requirements for Military Systems, Equipment and Facilities.
MIL-H-81581	-	Hose Assembly, Breathing Oxygen and Air
MIL-H-85047	-	Helmet Assembly, HGU-34/P.
MIL-C-85829	-	Coupling Assembly, Hose, Chemical, Biological, Radiological Protective.
MIL-V-85830	-	Ventilator, Tactical, Helicopter Aircrewman, Chemical, Biological, Radiological Protective.
MIL-M-85831	-	Mask, Helicopter Aircrewman, Chemical, Biological, Radiological Protective, Non-Oxygen.
MIL-I-85832	-	Intercom, Helicopter Aircrewman, Chemical, Biological, Radiological Protective Set.

STANDARDS

MILITARY

DOD-STD-100	-	Engineering Drawing Practices.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-143	-	Order of Precedence for the Selection of Standards and Specifications.
MIL-STD-461	-	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference.
MIL-STD-471	-	Maintainability Demonstration.
DOD-STD-480	-	Configuration Control - Engineering Changes, Deviations and Waivers.

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STANDARDS

MILITARY (cont'd)

MIL-STD-781	-	Reliability Tests Exponential
MIL-STD-875	-	Type Designation System for Aeronautical and Support Equipment.
MIL-STD-794	-	Procedures for Packaging and Packing of Parts and Equipment.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Chief of Naval Operations

OPNAVINST 3710.7	-	NATOPS General Flight and Operating Instructions; Promulgation of.

Naval Air Systems Command

NAVAIR 13-1-6.7	-	Aviation-Crew Systems Aircrew Personal Protective Equipment.
NAVAIR 13-1-6.10	-	Aviation-Crew Systems Aircrew Special Mission Equipment.
NAVAIR 01-110HCB-2 Series	-	NATOPS Flight Manual AH-1J Aircraft
NAVAIR 01-110HCB-1	-	NATOPS Flight Manual AH-1J Aircraft
NAVAIR 01-H1AAB-2 Series	-	NATOPS Flight Manual AH-1T Aircraft
NAVAIR 01-H1AAB-1	-	NATOPS Flight Manual AH-1T Aircraft
NAVAIR 01-110H1AAC Series	-	NATOPS Flight Manual AH-1W Aircraft
NAVAIR 01-H1AAC-1	-	NATOPS Flight Manual AH-1W Aircraft
NAVAIR 01-110HCE-2 Series	-	NATOPS Flight Manual UH-1N Aircraft

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Naval Air Systems Command (cont'd)

NAVAIR 01-110HCE-1	-	NATOPS Flight Manual UH-1N Aircraft
NAVAIR 01-250HDA Series	-	NATOPS Flight Manual CH-46A/D Aircraft
NAVAIR 01-250HDB-1	-	NATOPS Flight Manual CH-46A/D Aircraft
NAVAIR 01-250HDA Series	-	NATOPS Flight Manual CH-46E Aircraft
NAVAIR 01-250HDC-1	-	NATOPS Flight Manual CH-46E Aircraft
NAVAIR 01-230HMA Series	-	NATOPS Flight Manual CH-53A/D Aircraft
NAVAIR 01-230HMA-1	-	NATOPS Flight Manual CH-53A/D Aircraft
NAVAIR 01-H53BE Series	-	NATOPS Flight Manual CH-53E Aircraft
NAVAIR A1-H53BE-NFM-000	-	NATOPS Flight Manual CH-53E Aircraft
NAVAIR 01-230HLH-1	-	NATOPS Flight Manual SH-3A/G/H Aircraft
NAVAIR 01-230HLE-1	-	NATOPS Flight Manual SH-3D Aircraft
NAVAIR 01-230HLC-1	-	NATOPS Flight Manual HH/UH-3 Aircraft
NAVAIR A1-H60BB-NFM-000	-	NATOPS Flight Manual H-60 Aircraft
NAVAIR 01-260HCD-1	-	NATOPS Flight Manual H-2 Aircraft

DRAWINGS

NAVAIR 1605AS410	-	Ventilator Mount Assembly.
NAVAIR 1605AS100	-	Protective Assembly, Helicopter Aircrewman, Chemical, Biological, Radiological, A/P22P-9(V)

Naval Air Engineering Center

NAEC-ACEL-533	-	Anthropometry of Naval Aviators.

Headquarters, Department of the Army

Army Field Manual FM 3-5	-	NBC Decontamination.

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Headquarters, Department of the Army (cont'd)

U.S. Army Armament Munitions and Chemical Command Manual CRDEC CR-87033	-	Nuclear, Biological, and Chemical Contamination Survivability Methodology.

(Copies of specifications, standards, and other Governments documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI S12.6	-	Measurement of the Real-Ear Attenuation of Ear Protectors at Threshold, Method for.

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 4108	-	Test Methods for Thermal Protective Performance of Materials for Clothing by Open Flame Method.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AIR STANDARDIZATION COORDINATING COMMITTEE

AIR STD 61/14	-	Measurement of Protection Provided to the Respiratory Tract and Eyes Against NBC Agents in Particulate Aerosol, and Vapor Form.
AIR STD 61/23	-	Physiological Requirements for NBC Protection

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Protective assemblies furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for the opening of bids (see 4.3 and 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected as specified in MIL-STD-143.

3.4 Design and construction. The protective assembly (PASS) shall be designed and constructed as shown on Figure 1, consisting of three major components: mask, ventilator and intercom. The PASS shall also include a coupling assembly and a battery. The PASS shall provide head, eye and respiratory protection for the 3rd to 98th percentile male aircrewmember based on anthropometric data as specified in NAEC-ACEL-533 and the anthropometric matrix criteria sets of Table VI. Protection shall be provided from field concentrations of chemical and biological agents, toxins, in liquid, vapor and aerosol forms and radiological particles, while permitting the full performance of all phases of the helicopter mission profile without significant degradation in performance at ground level and in flight up to 10,000 feet. The PASS shall be self-sufficient for use when the aircrewmember is transient. The PASS shall not have protective performance degraded by exposure to perspiration, vomit, saliva nor any materials used in normal course of aircrewmember duties, including aircraft fuel, lubricants, de-icing, cleansing and decontamination agents and solutions. The design and construction of the system shall permit the use of the system after sanitization and decontamination using procedures of FM3-5 without damage or deterioration of the PASS or its protective capability. The procedures of CRDEC Manual CR-87033 shall be used as a guide in the design of the system to the maximum possible extent.

3.4.1 Materials. Materials used in the PASS shall conform to applicable specifications and standards and shall be compatible with environments specified herein. They shall be resistant to permeation by chemical and biological agents. The materials shall not pose a health hazard to the aircrew members. They shall be free from noxious or unpleasant odor, resistant to deterioration in use and storage, resistant to the adherence of radioactive dust, capable of providing protection against thermal effects while being fire resistant, not melting and not releasing toxic vapors. The materials shall be non-magnetic and when in contact with the skin, shall not cause any irritation.

3.4.1.1 Electrostatic compatibility. The PASS should be constructed of materials that do not produce or are susceptible to static electricity and shall be as specified in MIL-E-6051 and MIL-STD-461 for Class A1b.

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3.4.1.2 Protective treatment. Materials that are subject to deterioration when exposed to environmental conditions shall be protected against such deterioration in a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that might crack, chip or scale during normal service life under extremes of environmental conditions shall not be used.

3.4.1.3 Flamability. The materials used in the system shall be as self extinguishing and flame resistant as the A/P22P-9(V) Protective Assembly when tested as specified in 4.8.2.17.

3.4.1.4 Safety. The system shall meet the hazard and safety requirements of MIL-STD-1472.

3.4.2 Components. Each component shall be delivered completely assembled as specified in its specification. The components shall be capable of being connected and donned using connectors and accessories provided as part of each component. All system interfaces shall be functional for continuity of air flow or current flow, as applicable. All hose assemblies shall be as specified in MIL-H-81581.

3.4.2.1 Mask. The mask shall provide the aircrewman with filtered breathing air and protection against chemical, biological and radiological warfare agents on the head and neck areas. The mask shall also provide communication, valsalva, anti-suffocation, and a drinking capability while it is being worn. The design, construction and performance requirements of the mask and its components shall be as specified in MIL-M-85831.

3.4.2.2 Ventilator. The ventilator shall provide a continuous flow of filtered air to the PASS. The ventilator shall be a compact unit suitable for carriage on the person or mounted in the aircraft. The design, construction and performance requirements of the ventilator and its components shall be as specified in MIL-V-85830.

3.4.2.3 Intercom. The intercom shall be a small, lightweight, man mounted battery-powered intercommunication unit. The intercom shall provide the aircrewman wearing the PASS with the ability to converse comprehensively in CBR contaminated environments:

a. with another similarly dressed person.
b. with the wearer of a respirator with no microphone/telephone equipment.
c. with an unprotected person.

The design, construction and performance requirements of the intercom shall be as specified in MIL-I-85832.

3.4.2.4 Coupling assembly. The coupling assembly shall provide the aircrewman with the capability to readily separate the hose connecting the ventilator to the mask for emergency egress from the aircraft. The design shall allow operation with one chemical, biological, radiological (CBR) protective gloved hand and prevent inadvertent separation. The design, construction and performance requirements of the coupling and its components shall be as specified in MIL-C-85829.

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3.4.2.5 Battery. The battery shall provide power to operate the blower motor of the ventilator. The battery shall provide sufficient power to operate the ventilator for four hours over the range of mission and climatic conditions specified.

3.4.3 Interfaces. The PASS is intended for use in all USN/USMC helicopters (H-1, H-2, H-3, H-46, H-53 and H-60 Series) and shall be compatible with crew station hardware and the electrical power systems aboard the aircraft. It shall be compatible with all missions of these helicopters, including use of optical, targeting, navigation, communication and sighting devices. The PASS shall be compatible with laser protective equipment. The PASS shall be compatible and integrate with CBR protective clothing and devices in Mission Oriented Protective Postures (MOPP) as delineated in NAVAIR drawing 1605AS100. It shall also be compatible with all current aircrew life support equipment and emergency escape systems for use in flight, on the ground and in the water, including sequential use of the SRU-36/P Helicopter Emergency Escape Device (HEED) as required by OPNAVINST 3710.7, NAVAIR Manual 13-1-6.7, and NAVAIR Manual 13-1-6.10. Special attention shall be paid to the mask/flight helmet interfaces to provide maximum comfort and elimination of head pressure hot spots. The ventilator component shall be capable of being supported by the aircraft during flight with the standard mounting bracket (see NAVAIR 1605AS410) and operate either on battery power or 28 volt DC aircraft power when attached by power cord.

3.4.4 Human factors/physiological. Human performance and human engineering design criteria for the PASS shall be as specified in MIL-STD-1472. Physiological design criteria for the PASS shall be in accordance with ASCC AIR STD 61/23 or equal.

3.4.4.1 Wearability and comfort. The PASS shall be designed such that fully trained aircrew, not suffering from facial and head injuries, shall be capable of wearing the PASS comfortably while performing all phases of their assigned military duties while standing, ambulatory, or seated. The PASS shall not apply forces to the skin or facial muscles causing pressure points, skin irritation, abrasion, or edema. Design considerations shall include valsalva, drinking, anti-drowning, anti-suffocation, minimum weight and bulk, escape, mobility, and water survivability.

3.4.4.1.1 Weight and bulk. The PASS shall be as small and lightweight as possible, with negligible adverse affect on comfort and head mobility. The PASS shall conform to the following:

Total system weight - not greater than 12.0 pounds.

3.4.4.1.2 Center of gravity. The center of gravity of the head mounted portion of the PASS shall be as near to that of the bare head as possible.

3.4.4.1.3 Mobility. The PASS shall allow the aircrewman to have freedom of head movement to safely and effectively accomplish the mission of each specified aircraft as detailed in the applicable NATOPS Manuals for each specified aircraft.

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3.4.4.1.4 Fit. The PASS shall be sized to fit male aircrew within the 3rd to the 98th percentile for the anthropometric variables listed in Table VI "Matrix of PASS Accomodation Criteria Sets".

3.4.4.1.5 Ergonomics. The PASS shall be capable of being worn by aircrew safely and effectively performing mission required tasks for periods of up to 12 hours at light to moderate workrate (150-300 Kcal/hr) in temperate climate conditions (70 - 80°F WBGT)in all MOPP levels.

3.4.4.1.6 Airsickness. The PASS shall allow discharge of vomitus and have the capability to be resealed after discharge without subsequently removing part of the CBR ensemble.

3.4.4.2 Airflows. The operating PASS portable ventilator shall provide CBR agent-free positive hood ventilation and breathing. In the orinasal mask cavity, it shall be capable of meeting peak inspiratory and expiratory flows of 198 lpm and a total swing in pressure that is no greater than eight inches water gage (WG). Inspiratory and expiratory valves shall prevent a continuous flow of air through the mask.

3.4.5 Operating life. The PASS shall have an operating life of not less than 240 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be approved for use by the acquiring activity.

3.4.6 Storage life. The PASS and all components shall be designed to meet the performance and service life requirements of this specification after five years storage under worldwide climatic conditions specified in MIL-STD-210 when packaged in accordance with MIL-STD-794.

3.4.7 Interchangeability. All parts, subassemblies and assemblies having the same part number shall be identical as specified in DOD-STD-480 and DOD-STD-100. When any unit or part is interchanged, the PASS shall meet all performance requirements without adjustment or modification of any part or subassembly.

3.4.8 Nomenclature, nameplates and identification markings. Nomenclature and serial number assignment and nameplate approval shall be in accordance with MIL-N-18307. Type designation shall be in accordance with MIL-STD-875 as assigned by the Government. Parts and assemblies which are not assigned a Government type designation and equipment identification marking shall be marked in accordance with MIL-STD-130.

3.4.9 Workmanship. The PASS shall be uniform in quality and shall be free from irregularities, defects or foreign matter which could adversely affect safety, protection, performance, reliability or durability.

3.5 Performance.

3.5.1 Quantitative fit factor.

3.5.1.1 Normal mode.

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3.5.1.1.1 Chemical. For chemical protection, the Quantitative Fit Factor (QFF), with the ventilating flow 'ON' shall be not less than 10^4, when tested as specified in 4.8.2.1.1.

3.5.1.1.2 Biological. For biological aerosol not less than 0.002 percent penetration in the orinasal and lens cavities is permitted when tested as specified in 4.8.2.1.2.

3.5.1.2 Failure mode. For chemical protection the QFF with the ventilating flow 'OFF' shall not be less than 10^3. The protection factor shall be achieved continuously for not less than 30 minutes, when tested as specified in 4.8.2.1.3.

3.5.2 Human factors/physiological performance.

3.5.2.1 Respiratory performance.

3.5.2.1.1 Inhalation/exhalation resistance. With the ventilator blower motor 'OFF', the resistance for the PASS shall be as follows when tested as specified in 4.8.2.2:

a. Inhalation resistance - 2.0 in WG maximum at 3 cfm (250 Pa at 85 lpm).
b. Exhalation resistance - 1.0 in WG maximum at 3 cfm (100 Pa at 85 lpm)

3.5.2.1.2 Valsalva device. The PASS shall provide a means by which the wearer can occlude his nostrils in order to perform the valsalva maneuver to introduce air into the middle ear and paranasal sinuses during descent from altitude. In flight, it shall be possible to perform this maneuver with one hand while wearing CBR protective gloves when tested as specified in 4.8.2.3.

3.5.2.1.3 Anti-drowning. The PASS shall not have adverse flotation characteristics. The PASS shall not impede water survivability. It shall provide a quick disconnect to permit rapid egress and to avoid drowning when tested as specified in 4.8.2.4.

3.5.2.1.4 Anti-suffocation. The PASS shall provide for an anti-suffocation feature which shall be operable with one CBR gloved hand (or be capable of being emergency doffed in 15 seconds) when tested as specified in 4.8.2.5 and 4.8.2.13 .

3.5.2.2 Thermoregulation. The PASS shall be used as aircrew personal protective equipment in CBR contaminated environments which may be tropical, desert, or artic and be used onboard ship, in and around all types of USN/USMC aircraft.

3.5.2.2.1 Hyperthermia. The average body core temperature difference between the aircrewmember wearing the PASS (MOPP I and IV), and the aircrew wearing standard flight clothing, shall be not greater than 0.9°F (0.5°C). In MOPP I, II, III, and IV, body core temperature shall be not greater than 102°F (39°C) nor shall the difference between skin temperature/core temperature gradient be less than 0.9°F (0.5°C) at any time. Heart rate shall be not greater than 180 bpm during exercise or 140 bpm while at rest, when tested as specified in 4.8.2.6.1.

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3.5.2.2.2 Hypothermia/frostbite. During cold weather operations, the PASS shall provide adequate thermal protection to the wearer's head, face and neck. The average body core temperature of the aircrewmember wearing the PASS (MOPP I, II, III, or IV), shall be not less than 95°F (35°C) and skin temperature shall not drop low enough to allow frostbite to occur when tested as specified in 4.8.2.6.2.

3.5.2.3 Hydration. Drinking of water or liquid nourishment which is hot or cold via a drinking device shall provide a minimum flow 1 quart in 10 minutes. Drinking device preparation time shall be no more than 2 minutes and the start of water or liquid nourishment intake no more than 15 seconds after initiation of drinking, when tested as specified in 4.8.2.7.

3.5.2.4 Speech/communications. The PASS shall be fitted with an intercom system which will provide intercommunication on the ground and in flight, indoors and outdoors, and while the user is donning and doffing the CBR protective ensemble. The communications device shall provide the user with good auditory response from his immediate surroundings while leaving both hands free. The PASS intercom system shall be as specified in MIL-I-85832.

3.5.2.4.1 Noise. The level of noise transmitted to the microphone in the mask cavity from the PASS and the environment shall be not greater than the limits shown on Figure 2, when tested as specified in 4.8.2.8.

3.5.2.4.2 Sound attenuation. The PASS shall be subjected to the frequency spectrum specified in MIL-H-85047, Table I, and shall satisfy the requirements for sound attenuation specified therein, when tested as specified in 4.8.2.9.

3.5.2.4.3 Speech intelligibility. The PASS shall provide for intelligible voice communications as specified in MIL-H-46855, Modified Rhyme Test (MRT), and MIL-STD-1472. Normally acceptable intelligibility, 98 percent of sentences correctly heard, single digits understood, and scores of 91 or better shall be possible while wearing the PASS and helmet utilizing the PASS microphone, and under all other conditions specified in 4.8.2.10. Under high noise conditions (up to 115 dBA sound pressure level), a MRT score of 80 shall be acceptable when tested as specified in 4.8.2.10.

3.5.2.5 Vision/Optics. The PASS shall permit unobstructed and undistorted forward vision and optimum peripheral vision to the aircrew. The PASS shall be compatible with current helmet visor systems, all optical sighting devices in the applicable aircraft (see 3.4.3 Interfaces), and standard or specially designed spectacles. The PASS shall prevent build up of fog or ice without the use of anti-fogging compounds. Glare protection shall be provided by the in service helmet visor system. The systems shall be tested as specified in 4.8.2.11.1

3.5.2.6 Donning and doffing. The PASS worn in MOPP IV, and with other current life support equipment, shall be able to be donned or doffed in both normal and contaminated environments by a single person without hazard and without assistance. Donning shall be accomplished in less than 10 minutes. Doffing shall be accomplished in less than 10 minutes, when tested as specified in 4.8.2.12.

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3.5.2.6.1 Emergency doffing. The system shall be capable of being doffed in 15 seconds or less by an unaided aircrewman or have a design feature for anti-suffocation when tested as specified in 4.8.2.12.1

3.5.2.7 Flight and ground testing. The human factors performance of the PASS (MOPP I, II, III, IV) shall be verified during flight and ground testing. The PASS shall integrate with the aircrew ensemble as specified in OPNAVINST 3710.7, NAVAIR Manual 13-1-6.7 and NAVAIR Manual 13-1-6.10 and be comfortable during constant wear and not restrict head and body mobility when worn. As specified in 4.8.2.13, the aircrewmembers shall be capable of performing all of their missions while wearing the PASS in the applicable Mission Oriented Protective Posture of NAVAIR Drawing 1605AS100, as specified in the following manuals:

a. AH-1J NAVAIR 01-1IOHCB-2 Series
b. AH-1J NAVAIR 01-1IOHCB-1
c. AH-1T NAVAIR 01-H1AAB-2 Series
d. AH-1T NAVAIR 01-H1AAB-1
e. AH-1W NAVAIR 01-1IOH1AAC Series
f. AH-1W NAVAIR 01-H1AAC-1
g. UH-1N NAVAIR 01-1IOHCE-2 Series
h. UH-1N NAVAIR 01-1IOHCE-1
i. CH-46 A/D NAVAIR 01-250HDA Series
j. CH-46 A/D NAVAIR 01-250HDB-1
k. CH-46 E NAVAIR 01-250HDA Series
l. CH-46 E NAVAIR 01-250HDC-1
m. CH-53 A/D NAVAIR 01-230HMA Series
o. CH-53 A/D NAVAIR 01-230HMA-1
p. CH-53 E NAVAIR 01-H53BE Series
q. CH-53 E NAVAIR A1-H53BE-NFM-000
r. SH-3 A/G/H NAVAIR 01-230HLH-1
s. SH-3 D NAVAIR 01-230HLE-1
t. HH/UH-3 NAVAIR 01-230 HLC-1
u. H-60 NAVAIR A1-H60BB-NFM-000
v. H-2 NAVAIR 01-260HCD-1

3.5.2.7.1 Emergency escape. The PASS shall not hinder escape from an aircraft on the ground, following water ditching, or increase the risk of injury in the event of a crash and shall be as specified in 3.5.2.7.1.1 and 3.5.2.7.1.2.

3.5.2.7.1.1 Ground escape. The PASS shall not hinder escape from an aircraft on the ground. Maximum emergency egress time shall not be increased by more than 5 seconds average due to the PASS when tested as specified in 4.8.2.14.1.

3.5.2.7.1.2 Underwater escape. The PASS shall not hinder escape from an aircraft underwater. Maximum emergency egress time shall not be increased by more than 5 seconds average due to the PASS when tested as specified in 4.8.2.14.2.

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3.5.2.8 Canister replacement time. When tested as specified in 4.8.2.15, a single canister replacement time shall be 10 seconds or less. No special tools, materials or training shall be required. Degradation of protection shall be minimized.

3.5.2.9 Battery replacement time. When tested as specified in 4.8.2.16, battery replacement time shall be 30 seconds or less for each battery. No special tools, materials or training shall be required. No degradation of protection during battery change is permitted.

3.5.2.10 Transition. During transition, from portable mode to aircraft integrated mode, there shall be no loss in CBR protection. Maximum transition time from portable mode to aircraft integrated mode shall be not more than 15 seconds when tested as specified in 4.8.2.13.

3.5.2.11 Coupling Assembly. The coupling assembly shall provide the aircrewman with the capability to readily separate the hose connecting the ventilator to the mask for emergency egress from the aircraft. The design shall allow operation with one CBR gloved hand and prevent inadvertent separation. The design, construction and performance requirements of the coupling and its components shall be as specified in MIL-C-85829.

3.6 Maintainability. The need for routine servicing of the PASS and its components should be kept to an absolute minimum. The contractor supplying the equipment shall be responsible for the development and validation, of any special tools or test apparatus required for maintaining the efficiency of the PASS in service. Test equipment and tools already in service should be used as much as possible. Routine servicing should be possible in the presence of toxic agents. The PASS shall be capable of periodic functional checks over its service life. Design and construction shall be such as to minimize the retention of moisture and dust and permit operational stability, inspection, servicing, maintenance and cleaning.

3.6.1 Unscheduled maintenance - organizational level. The mean time to repair (MTTR) of the PASS, as demonstrated by MIL-STD-471, shall be less than or equal to 10 minutes at the organizational level. Operator maintenence includes normal care, cleaning, adjustment and battery replacement. Operator maintenance shall not require special tools or materials.

3.6.2 Unscheduled maintenance - intermediate level. The MTTR of the PASS, as demonstrated by MIL-STD-471, shall be less than or equal to 30 minutes at the intermediate level. Intermediate maintenance shall be performed using equipment developed for the PASS which shall be able to be operated by a qualified aircrew survival equipmentman.

3.7 Reliability. Reliability parameters for the PASS when tested as specified in MIL-STD-781 Test Plan IVC using 5 systems are as follows:

	Parameter	Threshold
(a)	Reliability (R_O)	0.95
(b)	MTBF(lower, single sided, 90 percent confidence level)	500.0 hrs.

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(c)	MTTR,(O level) scheduled or unscheduled	10.0	min.
(d)	MTTR,(I level) scheduled or unscheduled	30.0	min.
(e)	Operational availability (A_O)	0.95	
(f)	Operating life batteries (minimum) at -20°C	4.0	hrs.
(g)	System operational life (minimum)	240.0	hrs.

(R_O)= probability of completion of four hour mission without failure.
(A_O)= uptime/(uptime + downtime)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absense of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3)
b. First article inspection (see 4.4)
c. Quality conformance inspection (see 4.5)
d. Quality conformance verification inspection (see 4.6)

4.3 Qualification inspection. Qualification inspection shall be performed in the order specified in Table I, consisting of all the tests and examinations listed in this specification.

4.3.1 Qualification samples. Qualification samples shall consist of twenty protective assemblies consisting of at least one assembly of each size. Samples shall be forwarded to a test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (name) (date) for qualification inspection as specified in MIL-P-85833 and number under authorization (reference authorizing letter and number) (see 6.3).

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4.3.1.1 Disposition of samples. Upon completion of qualification inspection, all samples shall be consumed or destroyed and shall not be considered as part of the quantity to be delivered under contract.

4.3.2 Retention of qualification. To retain qualification, the contractor shall forward a report at six month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of a summary of the results of the tests performed, indicating the number of samples that have passed and the number that have failed during the six month period. If the summary of the test results indicates non conformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list. Failure to submit the report within 30 days after the end of each six month period may result in loss of qualification for the product. In addition to the periodic reports, the contractor shall immediately inform the qualifying activity of inspection data indicating failure of the qualified product to meet this specification. In the event that no production occured during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods, there has been no production, the manufacturer may be required at the discretion of the qualifying activity, to submit the qualified products to testing in accordance with qualifying inspection requirement and to state the reason for no production.

4.4 First article inspection. First article inspection shall be performed in the order specified in Table II.

4.4.1 First article samples. After award of the contract or order, the manufacturer shall submit at least 3 protective assemblies consisting of at least 1 of each size. The samples shall be representative of the construction, workmanship and materials to be used during production. When a manufacturer is in continuous production of these assemblies from contract to contract, submission of further first article samples on a new contract may be waived at the discretion of the acquiring activity (see 6.2.1). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1).

4.4.1.1 Disposition of samples. Upon completion of the first article inspection, one approved assembly shall be returned to the manufacturer for use in monitoring production. The other samples shall be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations and tests specified in Table III. The sampling and inspection levels shall be as specified in MIL-STD-105.

4.5.1 Inspection lot. An inspection lot size shall be expressed in units of PASS's made under essentially the same conditions and from the same materials and components and supplied by one manufacturer. The sample unit shall be one PASS.

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4.5.1.1 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing PASS's, fully prepared for delivery, made from essentially the same materials and components. The sample shall be one shipping container containing systems, fully prepared for delivery with the exeption that the shipping container need not be sealed.

4.5.2 Sampling for tests and examinations of PASS's. The sample size, acceptance criteria, tests and examinations required for the PASS shall be as specified in Table III.

4.6 Quality conformance verification inspection. At the option of the Government and upon completion of the quality conformance inspection requirements of 4.5, a random sample, one for every 150 or fraction thereof, shall be selected from each lot of systems (see 6.2.1). Each system selected as a sample unit, shall be forwarded to a laboratory designated at the time of the award (see 6.2.1). The designated laboratory shall conduct any or all the tests and examinations of 4.8 of this specification. The serial numbers of the units in the lot, represented by the sample units, shall be included with the data accom-panying the samples to the laboratory. The Government activity responsible for conducting the quality conformance verification program shall report the results of the tests and examinations to the designated inspection and acceptance activity specified in the acquisition document. Final acceptance of the lot from which the sample units were selected shall be based upon successful completion of the inspection program by the cognizant Quality Assurance Representative/Specialist at the contractor's facility; applying the appicable acceptance criteria specified in Table III.

4.6.1 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.7 Test conditions.

4.7.1 Equipment. Unless otherwise specified in the contract, all system tests shall be performed with all major components assembled together as specified in 3.4.2 forming the PASS.

4.7.2 Temperature and pressure. Unless otherwise specified in the contract, inspections shall be conducted at local ambient temperature and barometric pressure. Corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Inspection data provided by an instrument not calibrated to normal temperature and pressure (NTP) conditions shall be corrected to determine NTP conditions. NTP conditions are 70°F and 29.92 in. Hg.

4.8 Inspection methods.

4.8.1 Visual examination.

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...ified by the manufacturer shall certify that the materials and components

4.8.1.1 Protective assembly. Every PASS, selected as a sample unit from the lot, shall be visually examined for critical and minor defects to determine conformance to this specification. The classification of defects, (see Table IV), shall be used to classify the defects found.

4.8.1.2 Packaging. Each of the fully prepared shipping containers, containing PASS's, selected as a sample unit from the lot, shall be examined to determine that the packaging, packing and marking conform to this specification. The classification of defects, (see Table V), shall be used to classify the defects found.

4.8.2 Performance.

4.8.2.1 Quantitative fit factor. The Quantitative Fit Factor (QFF), shall be determined by quantitative leakage tests performed with a 10 person test panel, anthropometrically selected, to represent the 3rd to 98th percentile male aircrewman population in accordance with 3.4. The testing shall be both in a normal mode (with the ventilator motor operating) and in a failure mode (with the ventilator motor not operating). The test subjects shall wear the complete PASS and other applicable life support equipment. The tests shall be performed by an acquiring activity approved test facility where the test chamber is of adequate size to permit the simulation of all aircrewman movements during transition from a collective shelter to the aircraft, during flight operations and during transition back to the shelter.

4.8.2.1.1 Normal mode (chemical protection). As specified in 3.5.1.1.1, continuous measurements of penetration into the lens cavity and into the orinasal mask shall be recorded with instruments of sufficient accuracy to measure protection levels greater than 10^4. Penetration shall be measured using Sodium Chloride (NaCl), Dioctal Sebacate (DOS) or a suitable substitute. A test procedure shall require approval of the procuring activity prior to any testing. The test procedure shall be as specified in AIR STD 61/14 or equivalent.

4.8.2.1.2 Normal mode (biological protection). The QFF for normal mode biological protection shall be as specified in 3.5.1.1.2. Penetration shall be measured using a biological aerosol of B. globigii spores having a minimum concentration of 1×10^5 organisms/liter with a particle size of 1 to 5 microns diameter in accordance with the procedure outlined in Protection Branch Report of Test No. 9-66, Evaluation of E13R15 E13R16 Field Protective Masks, Physical Division, Fort Detrick, MD, 18 Feb. 66. Penetration will be measured on all sizes of mask properly fitted to clean shaven human test subjects wearing the PASS, CB clothing, and survival vest. Tests will be conducted under moderate activity.

4.8.2.1.3 Failure mode. The QFF for failure mode shall be as specified in 3.5.1.2.

4.8.2.2 Inhalation and exhalation resistance. Tests shall be conducted with a dry test meter and U-tube manometer. Air flow resistance shall be determined under NTP conditions at a flow rate of 3 cfm using (85 lpm) a Q-213 Inhalation/Exhalation Resistance Indicator or equal apparatus. The PASS shall meet the requirements as specified in 3.5.2.1.1.

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4.8.2.3 Valsalva device. The valsalva device shall be examined for full, and free operation and effectiveness. The valsalva device shall meet the requirements specified in 3.5.2.1.2.

4.8.2.4 Anti-drowning. During underwater escape trials of 4.8.2.14.2, the PASS shall be evaluated for flotation charateristics requirement of 3.5.2.1.3.

4.8.2.5 Anti-suffocation device. The anti-suffocation device shall be examined for full and free operation using one hand as specified in 3.5.2.1.4.

4.8.2.6 Thermoregulation.

4.8.2.6.1 Hyperthermia. At least 10 acclimatized test subjects wearing the PASS (MOPP I, II, III, IV) shall be tested in an environmental chamber. The test shall be conducted in two different environments; (a) Normal, 72°F and 50% relative humidity and (b) Hot-humid, 95°F and 60% relative humidity, with no wind. The subjects shall walk on a treadmill at a set speed of 3.0 MPH, 0 percent elevation. This workrate is representative of 3.3 METs (256 kcal/hr for a 50th percentile male weighing 171.61 lbs. as specified in NAEC-ACEL-533). Test subjects will walk for 25 minutes, followed by 5 minutes of rest. This cycle shall be repeated for four hours in the normal environment and for two hours in the hot-humid environment. Rectal temperature, at least 3 skin temperatures, and heart rate shall be recorded and monitored continuously. The test shall be terminated if any of the following occur: rectal temperature exceeds 102°F (39°C); skin temperature and core temperature merge within 0.9°F (0.5°C); heart rate is greater than 180 bpm during work or 140 bpm during rest; nausea; syncope; dry; hot skin; headache; subjective distress of the subject. The system shall be as specified in 3.5.2.2.1.

4.8.2.6.2 Hypothermia/frostbite. Not less than 10 test subjects wearing the PASS (MOPP I, II, III, IV) shall be tested in an environmental chamber at the following checkpoints: 0°C, -5°C, -10°C, -20°C, -30°C and -40°C. The chamber humidity and ventilation shall be recorded throughout the test. The subjects will be seated in helicopter fixed seat equivalents and be required to perform for up to 6 hours, a repeated 21 minute activity cycle consisting of 7 minutes of rest, 7 minutes of psychomotor tasks, and 7 minutes of pedaling a bicycle ergometer at 30. Rectal temperature, skin temperature, and heart rate will be monitored continuously and recorded not less than every five minutes. Frostbite shall not be allowed to occur; eleven surface skin temperatures will be monitored continuously. The test shall be terminated if any of the following occur: rectal temperature drops to 95°F (35°C), hand/digit temperature drops to 50°F (10°C), foot/toe temperature drops to 32°F (0°C), abnormal electrocardiograph as determined by Flight Surgeon or medical observer, the subject desires to terminate the test. The system shall be as specified in 3.5.2.2.2.

4.8.2.7 Hydration. 10 test subjects shall be used to measure the flow rate and the device preparation time. Preparation time shall be the time required for a subject to detach the drink tube cap from the mask and properly insert the device into the mask and into an M-17 canteen drinking cap. The

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mask shall be worn on the subject and the canteen in its carrier at the start. The drinking device shall be within an open carrying case. The test shall be conducted with subjects wearing the entire CBR ensemble. The mask shall pass the requirements as specified in 3.5.2.3.

4.8.2.8 Noise. External noise shall be simulated by a white noise emitted at an overall level of 110 dB. (from 10 to 20,000 Hz.) from a suitable loudspeaker located four feet away from and facing the PASS. The PASS shall be mounted on a dummy head. The ventilator shall be operating at an airflow of 100 lpm. The mask inhalation valve shall be open. The PASS shall meet the requirements as specified in 3.5.2.4.1.

4.8.2.9 Sound attenuation. The sound attenuation of the PASS shall be measured as specified in ANSI S12.6. The PASS, appropriately sized, shall be tested with SPH or HGU series flight helmet in place on ten anthropometrically chosen test subjects. The communications cord opening in the earcup grommet shall be plugged with RTV compound. Each subject shall be fitted with earcup pads as necessary to obtain a snug fit without being uncomfortable. Each subject shall use a new fitting pad set and earseals as required. The PASS shall meet the requirements as specified in 3.5.2.4.2.

4.8.2.10 Speech intelligibility. The Modified Rhyme Test as specified in MIL-STD-1472, shall be conducted with 10 test subjects (both speaker and listener). The test shall be conducted with protective clothing equipment, and in a representative aircrew ground environment containing the transmitting/receiving devices of the PASS. The PASS shall meet the requirements as specified in 3.5.2.4.3.

4.8.2.11 Vision/optics.

4.8.2.11.1 Anti-fogging. Not less than 10 test subjects wearing the PASS (MOPP I, II, III, IV) shall be tested in an environmental chamber at the following checkpoints: +5°C, 0°C, -5°C, -10°C, -20°C, -30°C, and -40°C. Test subjects shall perform one 30 minute session consisting of 10 minutes of rest, 10 minutes of Psychomotor tasks, and 10 minutes of pedaling a bicycle ergometer at 30 watts. They shall exhale into the PASS mask without condensation fogging the PASS eyelens.

4.8.2.12 Donning/doffing. Ten aircrewmen representative of the 3rd to 98th percentile male aircrewman shall don, then doff the PASS and all current life support equipment and CBR clothing. The PASS shall be as specified in 3.5.2.6.

4.8.2.12.1 Emergency doffing. Emergency doffing or operation of the anti-suffocation device shall be demonstrated by ground trials using 3 aircrewmen. The system shall be as specified in 3.5.2.6.1.

4.8.2.13 Flight and ground testing. Flight testing trials in H-1, H-2, H-3, H-46, H-53 and H-60 series aircraft and in trainers shall be conducted utilizing 10 aircrewmen representative of the 3rd to 98th percentile male aircrewman qualified to operate aircraft systems and equipment. Testing shall

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be conducted with the test subjects wearing the PASS in MOPP IV. Testing shall also evaluate valsalva, head mobility, body mobility and constant wear for 12 hours. In addition to all mission requirements the following performance shall be evaluated:

a. Interoperability
b. Interfaces
c. Clothing and equipment compatibility
d. Transition time
e. Communications
f. Wearability and comfort
g. Emergency doffing or alternate anti-suffocation device
h. Valsalva

The PASS shall meet the requirements specified in 3.5.2.7.

4.8.2.14 Emergency escape.

4.8.2.14.1 Ground escape. Ground escape trials shall be conducted in H1, H-1, H-2, H-3, H-46, H-53, and H-60 series aircraft or simulators utilizing 10 aircrewmen qualified to operate the aircraft systems and equipment. Testing shall be conducted with the aircrewmen wearing the PASS. The subjects shall perform emergency ground escape procedures following the appropriate NATOPS procedures for each aircraft and crew positions. The PASS shall be as specified in of 3.5.2.7.2 for the average of all aircrewmen.

4.8.2.14.2 Underwater escape. Underwater escape trials shall be conducted by 10 aircrewmen wearing the PASS in the Universal Multi-Placed Underwater Egress Trainer (9D5A) using the standard syllabus. The PASS shall be as specified in of 3.5.2.7.3 for the average of all aircremen.

4.8.2.15 Canister replacement time. The replacement time shall be the time it takes a subject to disconnect the canister(s) from the ventilator and replace each of the canister(s) with a new canister while the subject is wearing the PASS. The replacement canister(s) shall be unpackaged and the replacement operation shall be done at NTP conditions and with the subject wearing CBR gloves. The replacement time shall be as specified in 3.5.2.8.

4.8.2.16 Battery replacement time. The replacement time shall be the time it takes a subject to remove battery(ies) from the PASS and replace with new battery(ies) while the subject is wearing the PASS. The replacement battery(ies) shall be unpackaged and the replacement operation shall be done at NTP conditions with the subject wearing CBR gloves. The replacement time shall be as specified in 3.5.2.9.

4.8.2.17. Flammability. The Thermal Procedure Performance (TPP) rating shall be determined in the PASS system materials as described in ASTM D4108. The after flame, glow time and char length of the material shall be less than or equal to that of the current A/P22P-9(v) material tested at the same time.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial (see 6.2.1).

5.1.1 Level A. Each component of the protective assembly shall be preserved and unit packed as indicated in section 5 of the component's specification.

5.1.2 Commercial. The components of the protective assembly shall be preserved and unit packed as specified with ASTM D 3951.

5.2 Packing. Packing shall be Level A, Level B or Commercial (see 6.2.1).

5.2.1 Level A. The components preserved as in 5.1.1 shall be packed in a container conforming to MIL-STD-794, Table I as specified for Level A.

5.2.2 Level B. The components preserved as in 5.2.1 shall be packed in a container as specified in MIL-STD-794, Table I as specified for Level B.

5.2.3 Commercial. The components of the protective assembly shall be preserved and unit packed and then packed as specified in ASTM D 3951.

5.3 Marking. The unit packs and shipping containers of the protective assembly shall be marked and identified as specified in Appendix F of MIL-STD-129.

6. NOTES.

6.1 Intended use. The PASS covered by this specification is intended for use by aircrewmen to provide filtered breathing air and continuous protection, on the ground and in flight, to the eyes, skin of the head and neck and respiratory system against CBR agents in particulate, liquid, vapor and aerosol forms. The PASS is intended for use in all USN/USMC helicopters, specifically the H-1, H-2, H-3, H-46, H-53 and H-60 series.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Applicable Government part number.
c. Whether first article inspection is waived or the name and address of the first article inspection laboratory and the name of the Government activity responsible for conducting the first article inspection program (see 4.4.1).
d. If quality conformance verification inspection is required, the name and address of the designated laboratory and the Goverment activity responsible for verification program (see 4.6).
e. Selection of applicable levels of preservation (see 5.1) and packing (see 5.2)

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements. Manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they might be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command (Code 51122), Department of the Navy, Washington, DC 20361-5110; however, authorization for qualification of products shall be obtained from the Commander, Naval Air Development Center, Warminster, PA 18974, (Code 6031). Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of three complete PASS systems. The contracting officer should include specific instructions in acquisition documents, regarding arrangements for examinations, tests and approval of the first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Quantitative fit factor. The quantitative fit factor (QFF) is defined as the ratio of the concentration of agent in the immediate external environment of the PASS to the concentration of the agent in the specified compartment of the PASS:

$$QFF = \frac{\text{External agent concentration}}{\text{Internal agent concentration}}$$

6.5.2 MET. MET is a number expressing the ratio of a person's metabolic rate during an activity to his resting metabolic rate. Resting metabolic rate is approximately 3.5 ml of O_2 consumption per minute/kilogram body weight. One MET is also equal to 1kcal/kg" hr. Thus, the METs required for a given exercise load are calculated as follows:

$$METs = \frac{O_2 \text{ consumed during an activity}}{O_2 \text{ required for rest}}$$

$$METs = \frac{O_2 \text{ Consumed during activity}}{3.5 ml/kg/min}$$

Since 1 MET = 1kcal/kg" hr you can multiply the MET value by the body weight of the subject in kgc to obtain kcal/hr.

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6.6 Subject term (key word) listing.

Agent resistant
Airflow
Anti-drowning
Battery
Biological
Chemical
Coupling
Decontamination
Helicopter aircrewman
Intercom
Mask
PASS
Protection factor
Radiological
Survivability
Toxic
Valsalva
Vapor
Ventilator

Preparing activity:
Navy - AS
(Project 4240-N881)

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Table I. Qualification inspection.

Inspection	Requirement	Test method
Visual	---	4.8.1
QFF, chemical normal mode	3.5.1.1.1	4.8.2.1.1
QFF, biological normal mode	3.5.1.1.2	4.8.2.1.2
QFF, failure mode	3.5.1.2	4.8.2.1.3
Inhalation/exhalation resistance	3.5.2.1.1	4.8.2.2
Valsalva device	3.5.2.1.2	4.8.2.3
Anti drowning feature	3.5.2.1.3	4.8.2.4
Anti suffocation feature	3.5.2.1.4	4.8.2.5
Hyperthermia	3.5.2.2.1	4.8.2.6.1
Hypothermia/frostbite	3.5.2.2.2	4.8.2.6.2
Hydration	3.5.2.3	4.8.2.7
Noise	3.5.2.4.1	4.8.2.8
Sound attenuation	3.5.2.4.2	4.8.2.9
Speech intelligibility	3.5.2.4.3	4.8.2.10
Vision/Optics	3.5.2.5	4.8.2.11.1
Donning/Doffing	3.5.2.6	4.8.2.12
Emergency doffing	3.5.2.6.1	4.8.2.12.1
Flight testing	3.5.2.7	4.8.2.13
Ground escape	3.5.2.7.1.1	4.8.2.14.1
Underwater escape	3.5.2.7.1.2	4.8.2.14.2
Canister replacement time	3.5.2.8	4.8.2.15
Battery replacement time	3.5.2.9	4.8.2.16
Transition	3.5.2.10	4.8.2.13
Flamability	3.4.1.3	4.8.2.17
Weight	3.4.4.1.1	---

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Table II. First article inspection.

Inspection	Requirement	Test method	Sample number 1 2 3
Visual	-	4.8.1	X X X
Inhalation and exhalation resistance	3.5.2.1.1	4.8.2.2	X X X
Weight	3.4.4.1.1	-	X X X

Table III. Sample size, acceptance criteria, tests and examinations of the PASS.

Inspection	Section	Sample size	Acceptance criteria
Visual	4.8.1	MIL-STD-105 Normal level II S	AQL for major 1.0% AQL for minor 2.5%
Inhalation and exhalation resistance	4.8.2.2	MIL-STD-105 Normal level II S	AQL for major 1.0%

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Table IV. Classification of defects for visual examination of the PASS.

Critical	Minor
1. Material imperfections-foreign matter embedded.	201. Marking missing, insufficient, incorrect, illegible or not permanent.
2. Surface unclean, rough, mis-aligned, or containing cracks, nicks or other flaws.	
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or otherwise not securly retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

Table V. List of defects for packaging.

Item	Defects
Exterior and interior markings.	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials.	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship.	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strapping, etc.; bulging or distortion of the containers.
Exterior and interior weight or content.	Number per container is more or less than required; gross or net weight exceeds the requirements.

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Table VI. Matrix of protective assembly (PASS) accomodation criteria sets.
Criteria set dimensions*

Anthropometric Variables	Proportionate			Disproportionate		
	Set 1	Set 2	Set 3	Set 4	Set 5	Set 6
a. Head Length	7.32/3rd	7.8/50th	8.36/98th	7.54/15th	8.14/90th	7.67/30th
b. Head Circumference	21.6/3rd	22.7/50th	23.8/98th	22.19/20th	23.02/75th	22.7/50th
c. Menton-Vertex Height	7.95/3rd	8.73/50th	9.54/98th	8.32/15th	9.4/95th	8.52/30th
d. Menton-Sellon Length**	4.14/3rd	4.9/50th	5.63/98th	4.6/20th	5.35/90th	4.37/10th
e. Head Height (Tragion-Vertex)	4.66/3rd	5.16/50th	5.69/98th	5.0/25th	5.4/85th	5.25/65th
f. Nose Length	2.31/3rd	2.77/50th	3.23/98th	2.77/50th	3.14/95th	2.61/25th
g. Nose Protrusion	.72/3rd	.9/50th	1.12/98th	.82/20th	.90/50th	.74/5th
h. Head Breadth	5.74/3rd	6.13/50th	6.56/98th	6.24/70th	5.99/25th	6.02/30th
i. Bizygomatic Diameter	4.8/3rd	5.4/52nd	5.9/98th	5.45/60th	5.13/20th	5.45/60th
j. Bitragion-Menton Arc	11.73/3rd	12.7/50th	13.7/98th	12.43/30th	12.93/70th	12.68/50th
k. Bigonial Diameter	4.12/3rd	4.67/50th	5.37/98th	4.58/40th	4.39/20th	4.87/70th

* Dimensions are in inches followed by respective percentile for that variable taken from the 1964 naval aviator population.

** Dimensions were determined by adding nose length and menton-subnasal distance.

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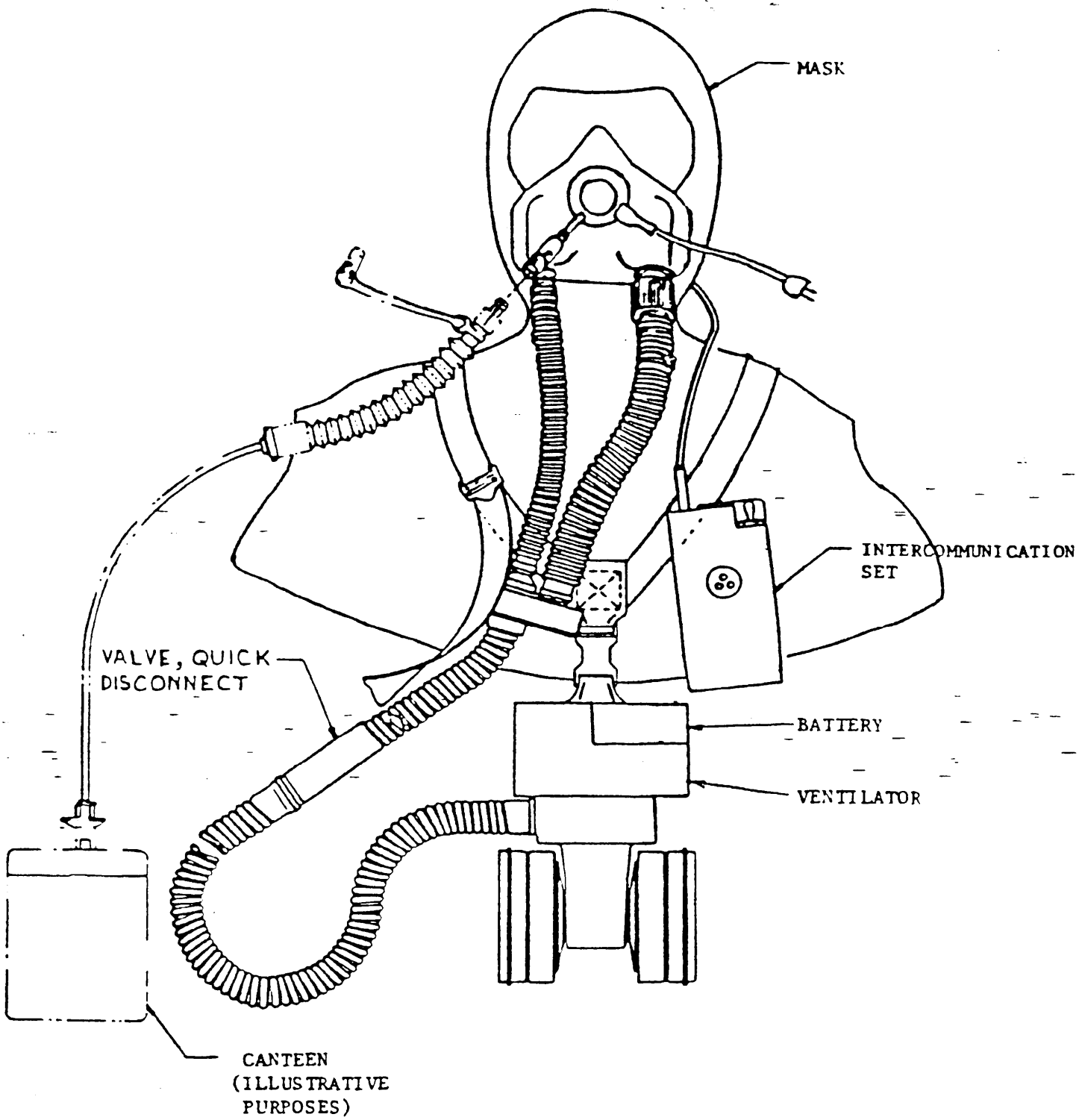


FIGURE 1. PROTECTIVE ASSEMBLY, CBR, HELICOPTER AIRCREWMEMBER

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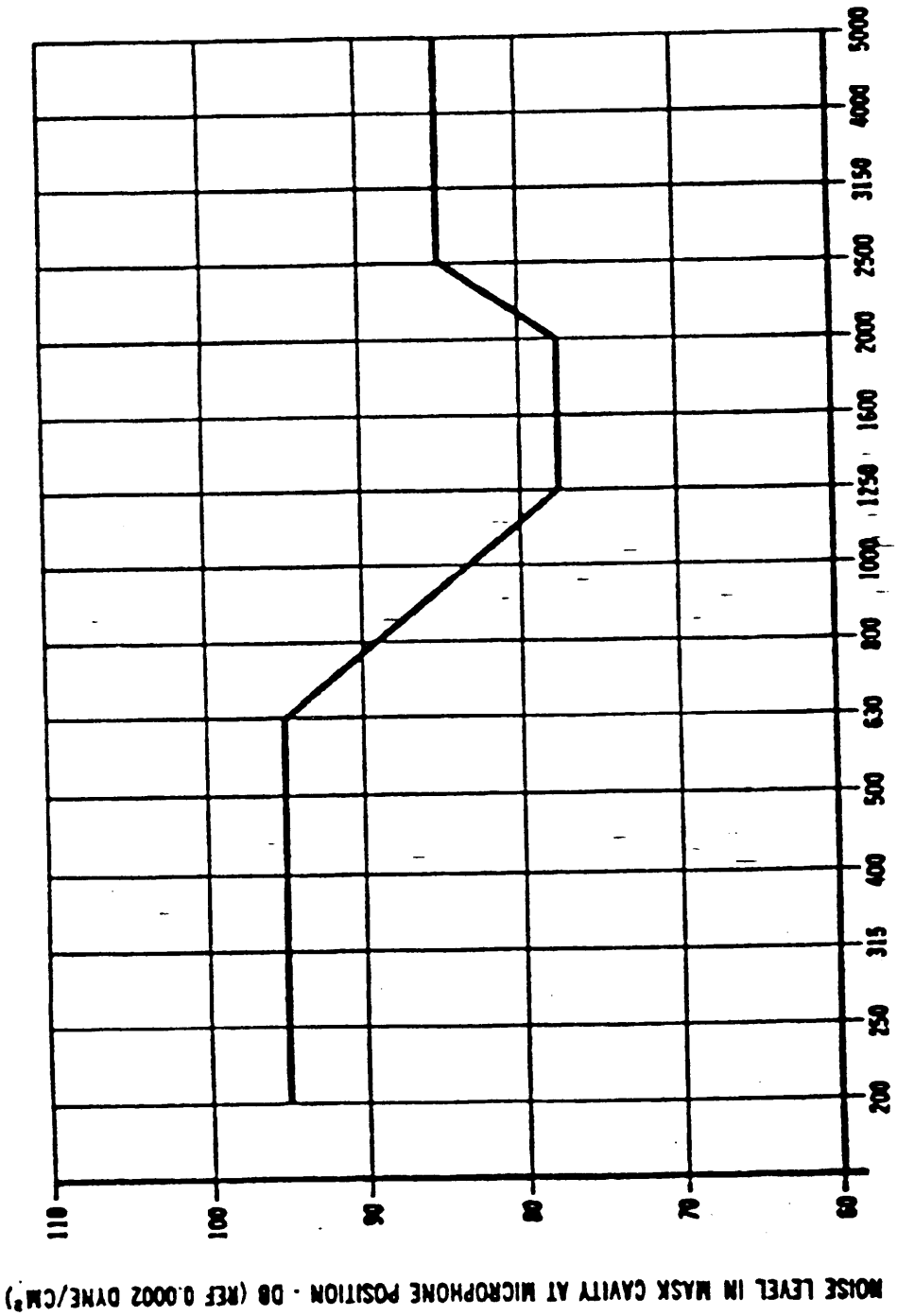


FIGURE 2. ORINASAL MASK CAVITY (MICROPHONE) NOISE LEVEL.

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